ROOT CLASSES - MATHEMATICS

Class 10th NCERT Study Material & Formulas

Chapter 5: Arithmetic Progressions (AP)

Key Concepts & Important Formulas:

* General Form of AP: a, a + d, a + 2d, a + 3d, ... where a is first term, d is common difference (d = a2 - a1).
* n-th term Formula: an = a + (n - 1)d.
* Sum of first n terms: Sn = n/2 * [2a + (n - 1)d] or Sn = n/2 * [a + l] where l = last term.
* Common difference can be positive, negative, or zero.

Solved Exam Examples & Practice:

Example: Find the 10th term of AP: 2, 7, 12, ...
a = 2, d = 7 - 2 = 5.
a10 = 2 + (10 - 1)*5 = 2 + 45 = 47.